12/10/24

Home Energy Savings Overview

The HES program will provide comprehensive and cost -effective retrofits for qualifying income eligible homeowner, partnering with local community organizations such as the Chicago Bungalow Association (CBA) and its sister organization, the Chicagoland Vintage Home Association (CVHA), as well as select Community Action Agencies and additional Energy Efficiency Service Providers (EESPs). These Retrofits will include:

- Comprehensive energy audit to identify weatherization and energy savings opportunities
- Comprehensive energy audit to identify weatherization opportunities
- Installation of weatherization (Wx) measures through a qualifying contractor
- Installation of a limited number of health and safety measures that would otherwise prevent the weatherization project from moving forward
- Installation or repair of mechanical systems on a highly limited basis

EESPs will be responsible for installation and onsite verification. QA/QC staff will conduct post-installation inspections through a sampling of in-field inspections and using data in the program database to minimize the overall impact to customers.

Notable Items:

Required Picture (RP) to be taken by EESP.

Regarding measure installation in a home:

- A homeowner is not able to “pick and choose” the measures that s/he wants installed in their home. Rather, the assessor will perform an energy audit and create a work order. Measures will be installed based on their cost-effectiveness, starting with the most cost-effective measures. Some measures that do qualify for a home may not be installed if there are more cost-effective measures ranked above it. It is important for the EESP to explain this to the homeowner.

Assessment Process

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1. Drive Up / Walk Up

   a) Identify House Type
      i) Housing Type Energy Profile
      ii) Common energy air sealing and insulation opportunities for housing type
      iii) Common challenges such as client comfort or structural issues for housing type
   b) Customer Introduction – See Appendix A.

2. External Inspection
   a) RP - Front of home with address
   b) RP – Any Pre-Existing Damage
   c) Roofing System Inspection
      i) RP – Roof overall condition
      ii) Determine and notate in Audit Data Collection Form:
         (1) Roofing Type – see Appendix B.
         (2) Roof Condition:
            (a) Good - All shingles are intact with NO warping/buckling
            (b) Average – Majority of shingles are intact with minimal warping/buckling
            (c) Poor – Missing shingles and signs of warping/buckling of those still intact.
      iii) Determine and notate quantities in Audit Data Collection Form:
         (1) Existing Ventilation – see Appendix C.
            (a) High Vents – Static or Ridge
            (b) Low Vents – Soffit
      iv) Determine presence and notate condition in Audit Data Collection Form:
         (1) Roofing System Components
            (a) Soffits
            (b) Chimney / Flashing
         (2) Bulk Water Mitigation
            (a) Gutters
            (b) Downspouts
            (c) Site drainage
   d) Insulation and Air Sealing Opportunities
      i) Walls - Determine and notate in Audit Data Collection Form:
         (1) Cladding type
         (2) Measurements
         (3) RP – 3 other sides of home

      ii) Foundations - Determine and notate in Audit Data Collection Form:
         (1) Type / types
         (2) Measurements (if multiple types)
         (3) If crawl space vents present notate quantity

iii) Measure any other structure components as needed (ASHRAE 62.2)

e) Cooling System - Determine presence and notate in Audit Data Collection
   i) Type
   ii) Size
   iii) Condition
   iv) RP – AC Condenser

3. Internal Inspection (Set House for diagnostics winter Conditions.)
   i) Attic - Determine and notate in Audit Data Collection Form – see Appendix D
      (1) Type / SQ FT / existing Rated R-Value
      (2) Air Sealing Opportunities
         (a) Bypasses
         (b) Duct Inventory / Condition
      (3) Health & Safety
         (a) Vermiculite
         (b) Gas Leaks
         (c) Knob & Tub
         (d) Exhaust Terminations
         (e) Baffles
         (f) Soffit chutes
         (g) Fire dam
   ii) RP Attic Photos:
      (1) Existing insulation
      (2) Major Bypasses
      (3) Baffle needs (soffit or fire dam)
      (4) Exhaust terminations
      (5) Hatch
      (6) Roof decking condition
      (7) Any pre-existing damage

b) Top Floor- Determine and notate in Audit Data Collection Form – see Appendix E
   i) Air Sealing Opportunities (Attic Access / Recessed Lights / Drop Soffits)
   ii) Insulation Opportunities knee wall / exterior wall
      (1) verification as needed borescope - wall plate method
      (2) Measurement as Needed (Window / door/ knee wall / hatch)
   iii) Exhaust fans (type / condition)
   iv) H&S
      (1) Gas Leak
      (2) Smoke and CO detectors

(3) Dryer Venting

v) RP – Top Floor Photos

(1) Insulation Opportunities (Exterior Wall / Knee Wall)

(2) Any pre-existing damage

c) Main Floor - Determine and notate in Audit Data Collection Form – see Appendix E

i) Air Sealing Opportunities (Exterior wall penetrations / fireplaces)

ii) Insulation Opportunities exterior wall

(1) verification as needed borescope / wall plate method

(2) Measurements as needed for windows and doors

iii) Exhaust fans (type / condition)

iv) H&S

(1) Gas Leak

(2) Smoke and CO detectors

(3) Dryer Venting

v) RP – Main Floor Photos

(1) Insulation Opportunities (Exterior Wall / Knee Wall)

(2) Any pre-existing damage

d) Foundation -Determine and notate in Audit Data Collection For – see Appendix F

i) Walls

(1) Type / SQFT /Existing R Value

(2) Air Sealing Opportunities

ii) Rim Joist

(1) Type / SQFT /Existing R Value

(2) Air Sealing Opportunities

iii) Heating System

(1) Type, size (BTUs)

(2) fuel type

(3) Condition

(4) H&S

iv) DHW System

(1) Type, size (BTUs)

(2) fuel type

(3) Condition

(4) H&S

v) Air Sealing Opportunities

(1) Bypasses

(2) Duct Inventory / Condition

vi) H&S

(1) Gas Leak

(2) Smoke and CO detectors

(3) Dryer Venting

vii) RP – Foundation Photos

(1) Insulation Opportunities (Exterior Wall/ Rim Joist /crawl wall ceiling)

(2) Any pre-existing damage

(3) FRN (with Venting)

(4) HWT (with Venting)

4. Pressure Diagnostics

a) Blower Door Test -

i) Flow chart - see Appendix G

b) Fan Flows Test

i) flow chart - see Appendix H

c) Determine Ventilation Requirements (ASHRAE 62.2) - see Appendix I

5. CAZ Zone Testing - see Appendix J

a) HDL

b) Spillage

c) Combustion Flue Analysis

i) Oven CO test - see Appendix K

6. Appendix A

Customer Introduction:

Hello, my name is _____a representative with the jointly funded Home Energy Savings Program. Today I will be conducting and energy assessment of your home to identify opportunities that will not only save you money on utility bills, but also increase your overall comfort within the home. As part of this process diagnostic tests will be conducted for Health and Safety purposes (gas leaks, high CO) as well as determining the home's relative "leakiness" in a numeric sense, and photos taken documenting the existing conditions.

Today's entire process will last approximately 2 – 2.5 hours.

First, I will conduct the External Inspection, where I will need access to all sides of the home, as well as the AC (if applicable).

Next, I will conduct the Internal Inspection, where I will begin in the Attic (if applicable) and work my way from the top of the home down. In that process, I will be identifying areas of the home that could become more energy efficient via insulation/air sealing.

I will then need access to test all combustion appliances, including your oven, heating system, and hot water tank. Afterward I will conduct air infiltration tests on any/all exhaust fans, and the entire home itself with a Blower Door installed within your (front) door.

Finally, we will review my findings and discuss next steps regarding the qualifying energy efficiency upgrades to be installed on your home. As well as answers any questions you may have.

7. Appendix B

Roofing Type:

- 3 – Tab Shingles
    - Three-tab shingles are distinguished by cutouts—tabs—made along their long lower edge. The result, that each shingle looks like three separate pieces when installed, but it's only one.

- Architecture Shingle
    - Architectural asphalt shingles contain no cutouts, but their lower portions are laminated with an additional asphalt layer. This creates the contoured, dimensional look that gives them their name. Asphalt sealant bonds the layers, reinforcing the shingles' waterproof capability.

- Flat Roof (most common types)
    - Bur or built up are often referred to as tar and gravel roofs. Layers of roofing felt, typically three feet wide are striped in and covered with melted hot tar. After the 2-4 piles or layers, roofs are often covered in aggregate or pea gravel. The small stones or gravel were used to protect the tar from the damaging effects of the suns UV rays.

- Modified Bitumen is similar to a built-up roof in that it is composed of asphalt rolls usually three foot wide, but a modified bitumen roof is usually only made of two piles or layers, a base sheet and a cap sheet. The cap sheet or top layer usually looks granulated like the granules on a shingle roof and often is light gray in color.

- EPDM Rubber roofs that are adhered are easy to spot, they are black in color and usually stretched tight over an insulation board or fiber board.

- Spray Polyurethane Foam roof begins with a layer of high-density polyurethane foam sprayed from a wand or sprayer. The foam is then coated with some type of waterproofing elastomeric topcoat. This topcoat often reflective gray or white in color.

- Thermoplastic roofing membrane are single layers of material usually white or gray but can be found in other colors. These membranes can be either adhered, similar to a EPDM rubber roof, or mechanically attached with screws and plates.

8. Appendix C

Attic Description/Location:_____

Ventilation Ratios

Attic w/vapor barrier: 1' FVA for every 300 sqft of attic surface area
Attic w/o vapor barrier: 1' FVA for every 150 sqft of attic surface area

FVA Conversions for Common Attic Venting Products	
Perforated vented soffit panels	Area x 13% = sq ft FVA
Gable or other louver style vents	Area x 50% = sq ft FVA
Various style roof vents	Area of hole cut in roof x 13% = sq ft FVA
Liqhtly restricted ridge vents	Area of cut in roof x 50% = sq ft FVA
Fabric type restricted ridqe vents	Area of cut in roof x 50% = sq ft FVA

Use manufacturer's FVA information whenever it is available. Reduce the FVA estimate by 50% for any products further restricted with window screen.

Attic Surface Area
(incl. walls and slopes that form ress/therm bounda

Component	Area
Total	

Existing Ventilation

Vent Location	FVA
Soffit	
Ridge	
Gable	
Roof	
Total	

Changes to Attic Ventilation Needed

Vent	FVA
Soffit	
Ridge	
Gable	
Roof	
Other:	
Total	

FVA Calculations

High (i.e. ridge or roofl

FVA Required:

FVA Present: FVA to Add: _____

Low *(i.e. soffit)*

FVA Required:

FVA Present: FVA to Add: _____

Attic Venting Diagram

9. Appendix D

Unfinished Attic

- Attic should be air sealed before it is insulated. NREL Standard Work Specifications (SWS) should be followed when air sealing.
- Unfinished attics should receive blown insulation, not batt insulation, when possible
- Attics overall R-value should be brought up to R-49
- Insulation should be blown to uniform depth
- Insulation markers should be installed every 300 square feet
- Cellulose is preferred insulation material
- Floored attics should be dense packed at 3.5lbs of cubic pressure unless homeowner has agreed to removing the flooring and blowing cellulose to R-49
- Attic hatches: blocking should be installed around the attic hatch, with plywood preferably
    - Hatch should be insulated to same R-value as attic
    - Weatherstrip should be installed on hatch to prevent air flow
    - Hatch must not be permanently sealed
    - For walk upstairs and pull-down hatches an insulated box should be built over stairs

Finished Attics

- Attic should be air sealed before it is insulated.
    - NREL Standard Work Specifications (SWS) should be followed when air sealing.
    - Floor-joist cavities under knee walls: Install rigid material between the joists; then foam or caulk the perimeter of each joist space. Alternatively, roll a fiberglass batt to fit tightly between each joist and cover with foam.

- Knee walls should be insulated to R-13+5 or R-20. R-13 of cavity insulation plus R-5 of continuous insulation with batt or rigid board
    - Hatch may be cut into knee wall(s) if needed
    - Hatch should be insulated to same R-value as attic

- Outer ceiling joists should be insulated to R-49, with open blow cellulose preferably
- Roof rafters should be insulated, if possible, with dense-packed cellulose at 3.5lbs of cubic pressure
- Collar beam should be insulated to R-49, if possible, with open blow cellulose

10. Appendix E

Exterior Walls

Exterior Walls should be insulated when possible

- Dense-pack cellulose is preferred material (for framed homes)
  - Walls can be insulated from exterior of the home (preferred) or interior of the home
  - Cellulose should be blow at a minimum of 3.50 pounds per cubic foot
- Injection foam is preferred material (for masonry foams)

11. Appendix F

Basement:

- Basement wall insulation may be installed, if is accessible (i.e., existing framing is in place to support insulation, NO GOLD PLATING OF HOMES such as finishing basement walls)
  - Rigid foam board or batt insulation may be used to bring total R – value to R10/13
    - "10/13" means *R*-10 continuous insulation on the interior or exterior of the home or *R*-13 cavity insulation on the interior of the basement wall.
- Rim joists should be insulated when accessible to bring total R-value to R10/13
  - Batt insulation is NOT allowed to be used for rim joist insulation

Crawlspace:

- Crawlspace Walls – before crawl space walls are insulated, moisture barrier needs to be installed on the ground, and any vents or ducts terminating inside the crawlspace need to be vented to the outside
  - Ground moisture barrier should be a minimum of 6 mil thick
  - Crawlspace walls should be insulated with rigid foam board or two-part spray foam to bring total R-value to R15/19

        - "15/19" means *R*-15 continuous insulation on the interior or exterior of the home or *R*-19 cavity insulation at the interior of the wall. Alternatively, compliance with "15/19" shall be *R*-13 cavity insulation on the interior of the basement wall plus *R*-5 continuous insulation on the interior or exterior of the home.
        - Fibrous insulation is not allowed to be used for crawlspace walls
- Crawlspace Ceiling - may be insulated, if it is determined that the floor is part of the thermal and pressure boundary
    - Insulation should be insulated to R-30
    - Insulation shall be installed to maintain permanent contact with the underside of the subfloor decking
    - As an alternative, the floor framing cavity insulation shall be in contact with top side of sheathing or continuous insulation installed on the bottom side of floor framing were combined with insulation that meets or exceeds the minimum wood frame R-value of R-20 or R-13+5 and that extends from bottom to the top of all perimeter floor framing members.

12.Appendix G

Blower Door Testing Checklist

Make sure there are no hazards that could be worsened by the test:

- ☐ Airborne Contaminants
- ☐ Mold and Mildew
- ☐ Asbestos-like material

Do no harm:

- ☐ Check for structural damage
- ☐ Assure customer that indoor climate will be within reason during the test.
- ☐ Move fragile items away from fan and potential leakage areas (like windows)

Place house under wintertime conditions:

- ☐ Close all exterior doors and windows; including storms
- ☐ Turn all combustion appliances *off* or to pilot
- ☐ Open all interior doors within the living space (this includes conditioned basements)
- ☐ Ensure that fireplaces/hearths are clean and have not been used in 24 hours. Cover opening with newspapers and tape
- ☐ Provide pressure relief for areas above drop ceilings. Always remove a ceiling tile before test.

Blower Door Testing

Blower Door Testing Reference Sheet

Set up the house for test - refer to checklist

Set up hoses and manometer gauges:

Details

Channel A (House Pressure): Insert hose into the REFERENCE tap and run end of hose outside, 5-6' out from blower door.
Channel B (Fan Pressure): Insert hose into the INPUT tap and insert end of hose into the tap on the fan.

Measure base pressure:

Details

Install all fan rings and the No-Flow Plate. Turn on manometer and press ***Baseline.*** Press ***Start*** and wait for pressure to stabilize before pressing ***Enter.*** Manometer should read at or near zero.

Select BD model and ring configuration

Details

Press MODE button twice to get to Pressure/Flow at 50 Mode

Take all rings out of the fan

Turn on fan and bring the house to -50 pa WRT outside:

Can you get the house to -50 pa?

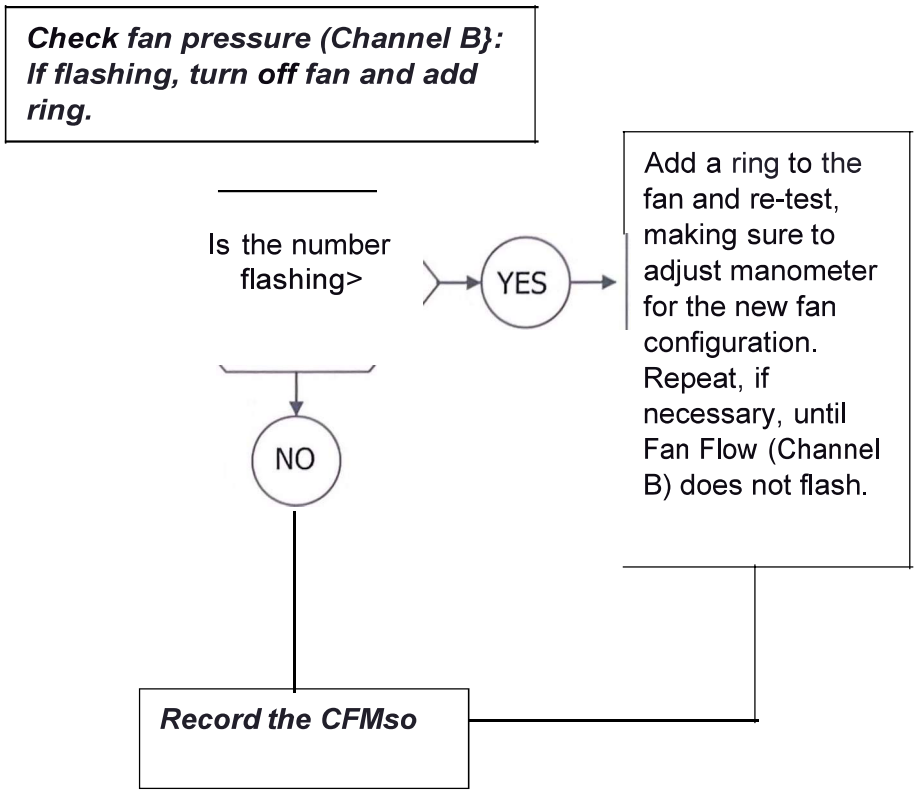
NO

Make sure all doors and windows are closed.

YES

Continue to next page

Blower Door Testing Reference Sheet {continued}



Cruise Control Operation for Blower Door Testing

1) Connect the manometer and blower door control with the communication cable.
2) Turn on the manometer and change the mode to measure pressure and flow at 50 pascals (PR/FL@S0). - The manometer will default to a blower door model 3 with an open fan configuration.
3) Turn the blower door control to the "on" position, but keep the fan speed turned all the way down.
4) Remove the No-Flow Plate and rings from the blower door.
5) Press "Begin Cruise" - The cruise target will default to 50 pascals.
6) Press "Start Fan" - The fan will start, and slowly increase in speed until the house is at -50 pascals or reaches the maximum cfm the fan can *move*.
7) If you can't reach -50 pascals, follow the steps outlined *above*.

13.Appendix H

Exhaust Meter Flow

1) Remove any covers or grilles and look for the manufacturer's CFM rating for the fan.
2) **Reinstall al l covers.**
3) Connect the Channel B input tap to the Exhaust Fan Flow Meter.
4) Use adjust the Exhaust Fan Flow Meter gate position and the **config settin gon the manometer.**

CONFIG	Flow M eter	flowCFM
A1	E1	44-124
B2	E2	21-59
C3	E3	10-28,

5) Turn on the manometer and select the proper settings:
   a. **MODE: PR/ FL**
   b. **DEVICE: ExH**
   c. **Config: As appropriate on the chart**
6) Turn on the fan and cover the grille with the Exhaust Fan Flow I\'1eter, appl}ting ste{ d}i **pressure.**
   a. If you are measuring a fan and get a "LO" indication, readjust the gate po : ion **and manometer configuration and repeat the test.**
   b. If you get a "LO" indication while measuring a fan at the E3 smallest gate setting,, **the measured flow is less than 10 CFM.**
7) **Record the installed flow rate of the fan.**

14.Appendix I

A ASHRAE Standard 62.2. Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings

DOE goals call for zero energy ready homes that are 50% more efficient than the 2009 IECC and whole-house retrofits that reduce energy use 25% in existing homes by 2025. By specifying minimum ventilation rates, ASHRAE 62.2 is a criticalenabling innovation that will contribute toDOE's long-term goal of saving the nation $2.2 trillion in energy-related costs through a 50% reduction in buildingenergy consumption.

Building America research and support were instrumental in developing and gaining adoption of ASHRAE 62.2, a residential ventilation standard that is critical to transforming the U.S. housing industry to high-performance homes.

High-performance homes provide more affordable, comfortable, healthy, durable, and safe homes for all Americans. That's a good thing. Tight construction is one of the most important cornerstones of high-performance homes, but is only possible with ensured dilution of indoor contaminants. And this ensured dilution is dependent on an effective base standard for whole-house and spot ventilation. This is why the ASHRAE
62.2 residential ventilation standard is critical to transforming the residential sector tohigh-performance homes. Building America research and support was instrumental indeveloping and gaining adoption of this vitally important standard.

To understand the importance of good indoor air quality, consider that people spend, onaverage, nearly 90% of their time indoors, with the majority of that time in their homes.And since concentrations of many pollutants indoors can be two to five times greater than in the outdoor air, ventilation is essential to maintaining good indoor air quality (Sherman 2004). However, historically residential ventilation was not considered a major concern by builders and home owners because it was believed operable windowsand envelope leakage provided enough outside air in their homes. But building envelopes have gotten progressively tighter as research conducted after the first oil crisis in the 1970s quantified the relationship between building air leakage and heat loss (and therefore energy loss). Today, more rigorous building codes and voluntary labeling programs are requiring builders to test for and meet stricter air leakage limits. For example, the 2009 International Energy Conservation Code, the first IECC code to specify a whole house airtightness limit, required homes to have air leakage of 7 or lessair changes per hour at 50 Pascals pressure (ACH 50) while the 2012 IECC requires homes to meet an airtightness limit of 3 ACH 50 or less in most climates (CZ3 and higher) and the Passive House program sets the limit at 0.6 ACH 50. ASHRAE 62.2 serves as a critical enabling technology as builders and codes approach zero energy homes with ever-tighter building enclosures.

These code changes would not have been possible without a recognized way to determine acceptable ventilation rates. Good indoor air quality is fundamental to

(top left) Building America-supported research and activities that were instrumental in getting the ASHRAE residential ventilation standard 62.2 written and adopted.

high-performance homes. Yet before ASHRAE 62.2, there was no generally accepted method for determining the ventilation rate a home needed to have for acceptable indoor air quality. As a result, there were fears that making homes too tight could result in significant indoor air quality problems which became a barrier to achieving airtight homes. In particular, builders were highly resistant to constructing tight homes and public programs often set weak air-tightening limits leaving too much energy efficiency on the table.

The establishment of ASHRAE Standard 62.2 resolved these issues by providing a minimum standard of care for the industry. The original standard, 62.2-2003 "Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings," was published by ASHRAE in late 2003. Since that time, there have continued to be revisions to the standard with new publications issued in 2004, 2007, 2010, and 2013.

ASHRAE 62.2 is of great benefit to the entire housing supply chain. Design and construction professionals now have a resource and guidance for specifying and integrating adequate ventilation in their projects. HVAC professionals and builders can reduce their business risk by employing an accepted set of criteria for whole-house and spot ventilation. The home sales process increasingly has a level playing field where the standard is enforced because it is no longer possible to gain a competitive first-cost advantage by leaving out adequate ventilation. And lastly, occupants have assurance that their homes are designed to provide adequate fresh air. (Sherman 2004).

Research and development sponsored by the U.S. Department of Energy (DOE) through its national laboratories and building research teams were instrumental in bringing about the creation and development of this nationally recognized standard for residential ventilation. In 1996, staff from Lawrence Berkeley National Laboratory (LBNL) urged ASHRAE to create a separate standard for residential ventilation for acceptable indoor air quality. ASHRAE had put forth Standard 62 in 1981, but it addressed ventilation in both commercial and residential buildings and was not specific to homes. From the committee's founding, Building America supported involvement from the national laboratories and research teams including staff from DOE, LBNL, Building Science Corporation, Florida Solar Energy Center, Washington State University Energy Office, and the Building Research Council at the University of Illinois.

Work by Building America-supported researchers from LBNL, Southface, the University of Illinois, FSEC, WSU, Davis Energy Group, and others was vital to getting the standard adopted by building organizations, including the Building Performance Institute (BPI), RESNET, the Air Conditioning Contractors of America (ACCA), California's Title 24 Energy Code, the U.S. Environmental Protection Agency's ENERGY STAR Certified

Homes Version 3.0, EPA Indoor AirPLUS, and Healthy Indoor Environment Protocols for Home Energy Upgrades, the DOE Weatherization Assistance Program (WAP), DOE's Workforce Guidelines for Home Energy Upgrades, and many "green" building programs. These efforts have resulted in the standard's use in hundreds of thousands of U.S. homes.

Building America continues to support ventilation research necessary to resolve issues of concern to the stakeholders, for example, LBNL's studies in range hood performance, diagnostics development, filtration, and smart ventilation; Building Science Corporation's field testing of ventilation strategies; and investigations on ventilation system performance by Pacific Northwest National Laboratory and others. As new construction methods and codes continue to push builders to construct tighter building enclosures, Building America research will further clarify issues regarding best practices for ventilation methods and rates in different climate zones. This research will shape future iterations of the standard.

ASHRAE 62.2 Core Elements

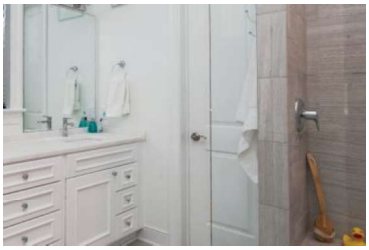
"Source Control" Exhaust Ventilation

KITCHEN

On-demand: 100 cfm -OR- Continuous: 5 ACH
12' x 12' x 8' → 100 cfm

FULL BATH

On-demand: 50 cfm -OR- Continuous: 20 cfm



"Whole Building" Mechanical Ventilation

$$Q_{fan} = 0.03A_{floor} + 7.5\,(BR + 1)$$

Floor area is surrogate for material sources
Bedrooms is surrogate for people, activities

↓

CREDIT FOR INFILTRATION

Allowances for intermittent ventilation, existing homes, and multi-family buildings

REFERENCES

ASHRAE Standard 62.2-2013. Ventilation for Acceptable Indoor Air Quality, https://www.ashrae.org/resources--publications/bookstore/standards-62-1--62-2

Sherman, M. 2004. ASHRAE's New Residential Ventilation Standard, ASHRAE Journal, Jan 2004, https://www.ashrae.org/File%20Library/docLib/Public/20031231103644_266.pdf.

Photos courtesy of Amerisips Homes LLC.

15.Appendix J

COMBUSTION APPLIANCE SAFETY INSPECTION FOR VENTED APPLIANCES*

*Vented appliances refer to natural draft appliances equipped with a barometric draft regulator or Category I appliances equipped with a draft hood or connected to a natural draft venting system.

The following combustion appliance safety inspection must be completed to determine if fossil fuel-fired appliances are operating safely under a depressurized condition.

MONITOR INDOOR AMBIENT CARBON MONOXIDE (CO)

Ambient CO must be monitored at all times during the test and actions taken as per the table below:

Required Actions in Response to Ambient CO Measurements (from ANSI/BPI-1200, Section 7.3.3.3)

70 ppm or greater	36 ppm-69 ppm	9 ppm- 35 ppm
• Terminate the inspection. • Notify the homeowner -occupant of the need for all building occupants to evacuate the building. • Leave the building and the appropriate emergency services shall be notified from outside the home.	• Advise the homeowner -occupant that elevated levels of ambient CO have been detected. • Open windows and doors. Recommend that all possible sources of CO be turned off immediately. • Where it appears that the source of CO is a permanently installed appliance, recommend that the appliance be turned off and advise homeowner – occupant to contact a qualified professional.	• Advise the homeowner - occupant that CO has been detected. • Recommend that all possible sources of CO be checked and windows and doors opened. • Where it appears that the source of CO is a permanently installed appliance, advise the homeowner - occupant to contact a qualified professional.

DEPRESSURIZE THE COMBUSTION APPLIANCE ZONE

Complete the following steps to place the CAZ under the greatest depressurization achievable given the weather/temperature conditions at the time of the inspection. Once you have determined that the greatest possible depressurization has been achieved, keep the CAZ in this depressurized state during all spillage assessment and CO measurements of vented appliances.

1. Place all combustion appliances located within the CAZ in their standby mode and prepare for operation.
2. Fires in woodstoves and/or fireplaces must be fully extinguished, with no hot coals or embers. Close fireplace dampers and any fireplace doors.
3. Close all building exterior doors and windows. Close all CAZ doors. Close the interior doors of all rooms except for rooms with an exhaust fan and rooms with a central forced air system return. Leave outdoor openings for combustion air open.
4. Turn off any mechanical ventilation and forced air cooling or heating system blowers.
5. Using a calibrated manometer or similar pressure measuring device intended for this purpose, **measure and record the baseline pressure in the CAZ with reference to (WRT) outside.** Compare this measurement with subsequent pressure measurements to determine the greatest negative pressure achievable in the CAZ.
6. Turn on the following exhaust equipment: clothes dryers (check and clean the dryer filter and look for blockage at the external vent damper prior to operation), range hoods, and other exhaust fans. If there are speed controls, operate the exhaust equipment at the highest speed setting. Do not operate a whole house cooling exhaust fan.
7. **Measure and record the pressure in the CAZ WRT outside.**
8. Turn on any central forced air system blowers and **measure and record the pressure in the CAZ WRT outside.**
   - If the pressure in the CAZ becomes more negative WRT outside after the blower is turned on, leave the blower ON during combustion appliance safety inspection.
   - If the pressure in the CAZ becomes more positive WRT outside after the blower is turned on, leave the central forced air system blowers OFF during the combustion appliance safety inspection.
9. Open interior door/s directly leading to the CAZ. **Measure and record the pressure in the CAZ WRT outside.**
   - If the pressure in the CAZ becomes more negative WRT outside after the door(s) are opened, leave the door(s) open during the combustion appliance safety inspection. *(Alternatively, pressure differential diagnostics may be used to determine proper door configuration to create the greatest CAZ depressurization. Pressure differential diagnostics may include manometer readings or a visual indicator, such as smoke.)*

SPILLAGE ASSESSMENT AND CO MEASUREMENT

Starting with the appliance with smallest BTUh input rating, follow lighting instructions and place in operation. Adjust the thermostat or control so the appliance will operate continuously.

In Cold Vent (Except Domestic Water Heaters)

Cold vent pertains to an appliance for which the heat setting is turned to OFF.

1. Assess spillage at **5 minutes** of main burner operation.
   - Action levels for spillage occurring at 5 minutes of main burner operation: See Table D.1.A.
2. Measure CO in undiluted flue gas at **5 minutes** of main burner operation. Compare the CO measurement with the appropriate CO threshold from Table 1, below.
   - Action levels for CO exceeding the appropriate threshold in Table 1: See Table D.1.B.

In Domestic Water Heaters or Warm Vent

Warm vent pertains to an appliance for which the heat setting is turned to ON.

1. Assess spillage at **2 minutes** of main burner operation.
   - Action levels for spillage occurring at 2 minutes of main burner operation: See Table D.1.A.
2. Measure CO in undiluted flue gas at **5 minutes** of main burner operation. Compare the CO measurement with the appropriate CO threshold from Table 1, below
   - Action levels for CO exceeding the appropriate threshold in Table 1: See Table D.1.B.

Multiple Combustion Appliances Sharing Chimney and/or Venting System

When a chimney and/or venting system is shared by multiple combustion appliances, use the following procedures to test combustion appliances for spillage and measure CO level in undiluted flue gases.

1. Test combustion appliances in order from lowest BTUh input rating to highest BTUh input rating.
2. Assess the appliance with the lowest BTUh input rating for spillage and CO measurement in undiluted flue gas.
   - If the vent is cold, assess spillage and measure CO at 5 minutes.
   - If the vent is warm (and/or if the appliance is domestic water heater) assess spillage at 2 minutes and measure CO at 5 minutes.
3. Upon completion of spillage testing and CO measurement of the first appliance, place the next largest BTUh combustion appliance in operation while the first appliance is still firing. Do not wait for the chimney to cool.
4. Retest the first appliance for spillage when the second appliance has reached 2 minutes of main burner operation. Test the second appliance for spillage immediately thereafter.
5. Measure CO level in the undiluted flue gas of the second appliance at 5 minutes of its main burner operation. Continue this process for each additional commonly-vented combustion appliance in order of BTUh input rating until all are running simultaneously.
6. Action levels for spillage on each appliance: See Table D.1.A
7. Compare the CO measurement with the appropriate CO threshold from Table 1.
   - Action levels for CO exceeding the appropriate threshold in Table 1: See Table D.1.B.

Table 1: CO Thresholds for Fossil Fuel-Fired Combustion Appliances (from ANSI/BPI-1200, Section 7.95)

Table 1
CO Thresholds for Fossil-Fuel Fired Combustion Appliances

Appliance	Threshold Limit
Central Furnace (all categories)	400 ppm air free[4]
Boiler	400 ppm air free
Floor Furnace	400 ppm air free
Gravity Furnace	400 ppm air free
Wall Furnace (BIV)	200 ppm air free
Wall Furnace (Direct Vent)	400 ppm air free
Vented Room Heater	200 ppm air free
Unvented Room Heater	200 ppm air free
Water Heater	200 ppm air free
Oven/Broiler	225 ppm as measured
Clothes Dryer	400 ppm air free
Refrigerator	25 ppm as measured
Gas Log (gas fireplace)	25 ppm as measured in vent
Gas Log (installed in wood burning fireplace)	400 ppm air free in firebox

Table D.1.A: Action Levels for Spillage in Combustion Appliances (from ANSI/BPI-1200, Annex D)

TABLE D.1.A ACTION LEVELS FOR SPILLAGE IN COMBUSTION APPLIANCES **The following actions shall be taken when spillage occurs under the specific circumstances detailed below.**	
TEST RESULT	ACTION REQUIRED
Greatest CAZ depressurization occurs with the air handler on*	Conduct further analysis of the distribution system to determine if leaky ducts or other HVAC-induced imbalances are the cause of the spillage. If so, recommend distribution system repairs that will reduce or eliminate the CAZ depressurization.
Greatest CAZ depressurization occurs with door to CAZ closed, but is alleviated when door to CAZ is open*	Recommend measures to improve air transfer between the CAZ and the core of the house
The cause of spillage has been traced to excessive exhaust** independent of CAZ door position, air handler, or a problem with the flue[t]	Verify that sufficient combustion air is available per *ANSI Z223.1/NFPA 54* for gas-fired appliances and *NFPA 31* for oil-fired appliances or recommend verification by a qualified professional and/or Recommend further evaluation/service by a qualified professional to address the venting/combustion air issues
*In the case where both spillage and excessive CO are present, in addition to the specific recommendations above, recommend that the appliance be shut down until it can be serviced by a qualified professional. ** Refers to exhaust caused by mechanical ventilation and/or other means of exfiltration. [t]When a recommendation to replace atmospherically-vented combustion equipment inside the pressure boundary is made, and when cost-effective, recommend replacement with direct-vented, or power-vented equipment (or non-combustion equipment, such as a heat pump), which is ENERGY STAR®-labeled.	

Table D.1.B: Action Levels for Carbon Monoxide in Combustion Appliances (from ANSI/BPI-1200, Annex D)

TABLE D.1 B. ACTION LEVELS FOR CO IN COMBUSTION APPLIANCES	
TEST RESULT	ACTION REQUIRED
Unacceptable CO level	Advise the homeowner/occupant that the appliance should be serviced immediately by a qualified professional Note: If ambient CO levels do not exceed 70 ppm, testing of other appliances and other audit procedures may continue at the discretion of the auditor
Acceptable CO level	No action required

NOTE: The Action Levels in Tables D.1.A and D.1.B above are based on the following criteria:

- CO measured at 5 minutes of main burner operation
- Spillage assessed at 2 minutes of main burner operation for warm vent
- Spillage assessed at 5 minutes of main burner operation for cold vent
- CO level at or below threshold in Table 1 for the appliance being tested is ACCEPTABLE
- CO level exceeding threshold in Table 1 for the appliance being tested is UNACCEPTABLE

TERMS AND DEFINITIONS

Barometric Draft Regulator	A balanced damper device attached to a chimney, vent connector, breeching, or flue gas manifold to control chimney draft.
Carbon Monoxide (CO) Air Free	Air free emission levels are based on a mathematical equation (involving carbon monoxide and oxygen or carbon dioxide readings) to convert an actual diluted flue gas carbon monoxide testing sample to an undiluted air free flue gas carbon monoxide level utilized in the appliance certification standards. For natural gas or LP gas, using as-measured CO ppm and O2 percentage: $$co_{AFppm} = \left(\frac{20.9}{20.9 - o_2}\right) X\ co_{ppm}$$ Where: CO_{AFppm} = Carbon monoxide, air-free ppm; CO_{ppm} = As-measured combustion gas carbon monoxide ppm; O_2 = Percentage of oxygen in combustion gas, as a percentage
Category I Vented Appliance	An appliance that operates with a *non-positive* vent static pressure and with a vent gas temperature that *avoids* excessive condensate production in the vent.
Category II Vented Appliance	An appliance that operates with a *non-positive* vent static pressure and with a vent gas temperature that *can cause* excessive condensate production in the vent.
Category III Vented Appliance	An appliance that operates with a *positive* vent static pressure and with a vent gas temperature that *avoids* excessive condensate production in the vent.
Category IV Vented Appliance	An appliance that operates with a *positive* vent static pressure and with a vent gas temperature that can cause excessive condensate production in the vent.
Combustion Appliance Zone (CAZ)	Room and enclosed air volume that contains a combustion appliance. This may include, but is not limited to, a mechanical room, mechanical closet, or main body of the house.
Common Vent	That portion of a vent or chimney system that conveys products of combustion from more than one appliance.
Direct Vent Appliance	An appliance that is constructed and installed so that all air for combustion is derived directly from the outdoors and all flue gases are discharged to the outdoors.
Draft Hood	A draft hood acts as a pressure break between the vent system and the appliance and eliminates stack action. Without the draft hood, the vent could experience excessive draft, flame instabilities, and possibly pilot outage.
Spillage	Entry of combustion products into a building from dilution air inlets, vent connector joints, induced draft fan case opening, combustion air inlets, or other locations in the combustion or venting system of a vented combustion appliance (boiler, fireplace, furnace, or water heater), caused by backdrafting, vent blockage, or leaks in the venting system.
Vent	A passageway used to convey flue gases from appliances or their vent connectors to the outdoors.

16.Appendix K

Evaluation of Range Burner Operation:

1. Calibrate the CO analyzer.

2. While monitoring ambient CO levels, light each individual burner. If ambient CO levels exceed 35 ppm during testing, turn off the burners and ventilate the space.

3. If the burners do not ignite properly or do not burn cleanly, a clean and tune of the appliance shall be recommended.

4. If the appliance is located in a confined space and mechanical ventilation is not readily available, mechanical v.ntilation shall be recommended with a minimum capacity of 25 cfm continuous or 100 cfm intermittent.

Oven Testing Procedure:

Remove items from the oven before testing.

1. Turn on range/oven exhaust or recirculating fan, and open a window.

2. Continue to evaluate ambient CO.

3. Turn the oven to 500 degrees. If ambient CO level exceeds 35 ppm, turn off the appliance and ventilate the space.

4. Measure the undiluted CO after 5 minutes of burner operation (if the appliance has reached "Steady State"). Steady state operation is indicated by a consistent undiluted CO measurement. Variances of more than 1 to 2 ppm are an indication that the appliance has not reached steady state operation.

S. If the appliance has not reached "Steady State", you may wait a maximum of 10 minutes before recording the undiluted CO measurement. Indicate with notes the measured range of undiluted CO if the appliance still has not achieved steady state operation.

6. Refer to Building Analyst Technical Standards for interpretation of test results and required action level.
Ambient CO before testing:_____ Highest Ambient CO during testing:___ _

Steady State Undiluted CO:_____

Required Action from BA Standards:_